

Planning support systems: an inventory of current practice

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Abstract

Planning support systems are a subset of computer-based geo-information instruments, each of which incorporates a unique suite of components that planners can utilise to explore and manage their particular activities. The components may include data sets, computer algorithms and display facilities, as well as more abstract theoretical constructs, knowledge and modelling capabilities. PSS are used to support the planning process by communicating information as well as by generating solutions. An Internet-based inventory was established in 2000–2001 to identify the existence and use of PSS in planning practice worldwide. In this paper we take a closer look at the instruments recorded in the inventory and some of their main characteristics. Some recommendations are made for the development and application of PSS in the future.

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1. Introduction

A plethora of methods and tools have been used by professional planners over many years to support their various planning activities in different contexts. Nevertheless, it can be argued that planning practitioners have never fully embraced the diversity of methods, techniques and models developed in the research laboratory to analyse spatial problems, evaluate alternative options or project future scenarios. Most geo-information tools do not readily fit the changing needs of the planning

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profession; they are far too generic, complex, inflexible, incompatible with most planning tasks, oriented towards technology rather than problems, and too focused on strict rationality, as documented by number of commentators including Couclelis (1989), Ottens (1990), Scholten and Stillwell (1990), Klosterman (1994), Worrall (1994), Bishop (1998), Nedovic-Budic (1998) and Geertman (1999).

Partly as a consequence of this mismatch between the demands of practitioners and the supply of methods and techniques, a new generation of novel or redeveloped tools to facilitate 'new' planning practice has materialised under a generic term—planning support systems (PSS). These systems include instruments relating to geo-information technology that have been primarily developed to support different aspects of the planning process, including problem diagnosis, data collection, mining and extraction, spatial and temporal analysis, data modelling, visualisation and display, scenario-building and projection, plan formulation and evaluation, report preparation, enhanced participation and collaborative decision-making. In the case of the latter—the development of support for participatory planning processes—a variety of new instruments has emerged ranging from electronic conference board rooms (group decision support systems) (Laurini, 1998) and GIS-supported collaborative decision-making tools (Nyerges & Jankowski 1997) to Web-based mediation systems for co-operative spatial planning (Gordon, Karacapilidis, Voss, & Zauke, 1997; Shiffer, 1992) and support tools for different planning tasks (Geertman, 2002a; Hopkins, 1998, 1999; Kammeier, 1999; Klosterman, 1999b; Singh, 1999).

Despite these developments, there remains a general lack of knowledge of and insight into the development, adoption and implementation of PSS in different planning contexts. The state-of-the-art in terms of the adoption of PSS and their real contribution in practise has remained uncertain. Therefore, in 2000 we decided to construct an inventory of PSS. This article provides insights into some of the important issues relating to the PSS contained in the inventory. It begins in Section 2 with the question of definition: what makes a PSS different from other geo-technology based systems like a geographical information system (GIS) and a spatial decision support system (SDSS)? The inventory, its context, aims and creation are discussed in Section 3. Thereafter, in Section 4, selected examples from the inventory are used to illustrate the range of applications that have been developed. Finally, Section 5 contains a number of recommendations.

2. Defining PSS

As indicated already, PSS consist of a wide diversity of geo-information tools that are dedicated to support public or private planning processes (or parts thereof) at any particular spatial scale and within a specific planning context. PSS have much in common with two other well-known systems: GIS and SDSS. So what makes a PSS different from a GIS or a SDSS?

PSS are generally regarded as systems in which technologies dedicated to the planning profession are brought together. Harris and Batty (1993) associated the

concept of PSS with combining a range of computer-based methods and models into an integrated system that is used to support a particular planning function. More precisely, in their opinion, a single PSS forms the framework in which three sets of components are combined: the specification of the planning tasks and problems at hand, including the assembly of data; the system models and methods that inform the planning process through analysis, prediction and prescription; and the transformation of basic data into information which in turn provides the driving force for modelling and design (through a cyclic process). In a similar vein, [Klosterman \(1997, 1999a\)](#) and [Brail and Klosterman \(2001\)](#) more recently, have described PSS as information technologies that are used specifically by planners to undertake their unique professional responsibilities. They suggest that PSS have matured into frameworks of integrated systems of information and software that synthesise the three components of traditional DSS—information, models, and visualization—and deliver them into the public realm.

In contrast, GIS can be regarded as general-purpose systems, comprising tools for handling geo-referenced data, and applicable for a wide diversity of tasks and problems in various settings. There are many generic GIS on the commercial market and systems such as MapInfo Professional, IDRISI or ArcGIS have now become commonplace on the desktop. On many occasions, a proprietary GIS will form part of a PSS, given the standard sets of functional capabilities that the former very usefully provides. SDSS, on the other hand, can be regarded as systems designed specifically to support a decision research process for complex spatial problems. They provide frameworks for integrating database management systems with analytical models, graphical display and tabular reporting capabilities, and with the expert knowledge of decision-makers ([Densham, 1991](#)). Although SDSS have much in common with PSS, PSS generally have to pay particular attention to long-range problems and strategic issues and may even be designed explicitly to facilitate group interaction and discussion. SDSS, on the other hand, are generally designed to support shorter-term policy-making by isolated individuals or by business organizations ([Clarke 1990, chap. 15](#)). Moreover, their prime dedication is towards supporting executive decision-making rather than professional planning activities.

In our view, the categories of PSS, GIS and SDSS are by no means mutually exclusive and have much in common, but each does possess certain distinguishing characteristics as indicated previously. PSS specifically support the whole of or some part of a unique professional planning task. They aid the planning process by providing integrated environments usually based on multiple technologies and a common interface. In some cases, the integrated environment is no more than a loosely coupled collection of computer-based tools. However, it is not so much the technology, with its capabilities and restrictions, that dictates the support function performed by the PSS, but the specific needs of the planning context in which the PSS is designed to operate. As a consequence, emphasis is focused on the first S (for support) in the PSS acronym in conjunction with the P, rather than on the second S (for systems), as has been the case in practice so often before. In summary, PSS distinguish themselves from GIS and SDSS by virtue of the fact that they are dedicated specifically to activities associated with planning in practice.

3. A PSS inventory

Stillwell, Geertman, and Openshaw (1999a) provide an assessment of planning practice at the beginning of the twenty-first century from which it can be concluded that the adoption of geo-information tools is far from widespread and from being effectively integrated into the planning process. Planners and designers remain at best distrustful, and at worst downright antagonistic, toward computer-based models (Harris, 1998, 1999). From these and other studies (e.g. Brail & Klosterman, 2001), it can be concluded no comprehensive picture of the extent of the development of PSS has yet been constructed and no assessment of the success of their adoption has been produced. As a result, different groups of researchers and developers may well be undertaking a substantial amount of work that duplicates or overlaps with work undertaken elsewhere. At the same time, the planning community as a whole has little idea of where to look when searching in the academic or planning literature for good examples of the use of PSS in practice, details of new instruments or advice and support for PSS. This is problematic for the potential consumers (planners) at a time when there is an increasing need for geo-information technology support. Moreover, the developers of PSS have to prove the worth of their products in real-world planning situations.

In view of these circumstances, a review of the field of PSS has been undertaken to identify innovative examples of geo-information technology-related applications and 'best practice' that fits into the PSS definition given above. A Web-based inventory of PSS was established in order to collect details of PSS implemented and operational in planning practice including those available as off-the-shelf commercial products and also including prototypes under development. Entries to the inventory (which can be found at <http://www.nexpri.nl>) were made between June 2000 and June 2001 and about 50 contributions were received from over 20 countries including the Netherlands, the UK, the USA, Italy, Columbia, Australia, New Zealand, Russia, South Africa, Malaysia, Sweden, Finland, the Czech Republic and Hong Kong.

Following careful scrutiny of the inventory entries, a number of systems from various countries covering a wide range of application areas of strategic support, land-use and infrastructure planning and environmental management, were selected for inclusion in more detailed form as chapters of a book (Geertman & Stillwell, 2003). In the next section, some of the PSS inventory will be described in summary.

4. Contents of the PSS inventory

Several conclusions can be drawn from this exercise of creating an inventory that provides insights into the concept of PSS and their application in practice. Firstly, the numbers of PSS are increasing world-wide. Whilst the inventory contained approximately contributions from several different countries across the world, the PSS selected for detailed specification in Geertman and Stillwell (2003) originate from 15 different countries. Most of the PSS appear to have been developed very recently, with only a small minority being over 5 years of age. As a consequence,

many of the PSS represented in the inventory are not mature, and some are still in an experimental or prototype stage of development. Certainly, the large majority of PSS are not available as proprietary, off-the-shelf systems.

A second important conclusion is that experience with PSS in actual planning practice is very limited. The real-world application of many systems seems to be confined to experimental trials, such as educational meetings with students or training sessions with professional planners, for instance. A small number of the systems have indeed matured into a stage of professional application. However, at the moment, the extent of these applications is limited and this makes it difficult to present robust conclusions and recommendations.

A third conclusion is that those systems recorded in the inventory reflect a very wide range of application areas. It is fruitful to examine this diversity in more detail, because this conveys the actual contribution that PSS are making and the potential of the systems for use in different areas of planning practice. The diversity is apparent in the aims, capabilities, content, structure and technology of the PSS, as well as in their appearance to users.

4.1. *Aims*

The PSS contained in the inventory have a wide diversity of aims. Some are dedicated to facilitate and/or enhance participation by the public and/or of stakeholders in the planning process; others can be characterised as tools dedicated to support specific tasks within planning processes like the uniform handling of building permits; others have been developed to inform the public about different planning and policy topics in their region or country; and others support specific forms of planning by practitioners such as strategic planning, land-use and infrastructure planning or environmental planning. Two very different PSS illustrate this diversity of aims in the category of environmental planning; one is an area-specific system known as the Planning System for Sustainable Development (PSSD) and the other is a task-specific system known as SPARTACUS.

The so-called PSSD is dedicated to support the tasks of professional planners in the Baltic Sea Region to enhance the sustainability of their policies (Hansen, 2001). Developed by a consortium of Finnish, Danish and German partners, it contains a wide diversity of instruments, put together on a web site, consisting of best practices, sustainability indicators, associated theories, scientific documents, supporting tools, meta-information, geo-data and dedicated methods. These instruments are interconnected, providing the opportunity to relate a specific planning question to some associated underlying theoretical notions that lead to the selection of dedicated methods, tools and data to handle the specific task in accordance with the (meta-) information provided. Fig. 1 illustrates an outcome of the PSSD, showing the result of an accessibility analysis within the Finnish region of Päijät-Häme. Accessibility is calculated by measuring travel time from each grid cell to the nearest of four different size categories of urban centres and then summarised. The result is a relative, and therefore comparable, measure of accessibility of urban centres, based on the principle of gravity modelling. The darker cells of the two urban centres in the south

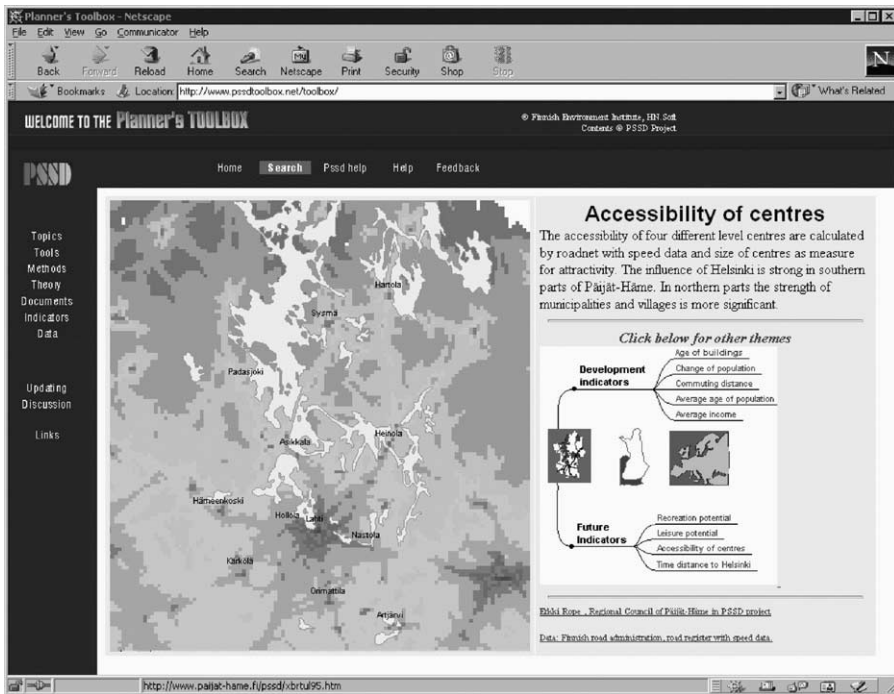


Fig. 1. PSSD interface showing the accessibility of centres near Helsinki.

of Fig. 1 indicate the relatively good accessibility via the road of these urban centres (<http://www.pssdtoolbox.net/plannertool.htm>).

In contrast, the SPARTACUS system (System for Planning and Research in Towns and Cities for Urban Sustainability) is more task-specific and has been developed by another consortium of partners from Finland, the UK, Spain and Germany. It is dedicated to support sustainable urban policies at different locations and it consists of a land use/transport model (MEPLAN) combined with a set of urban sustainability indicators, a GIS-based Raster method to calculate values for some of the indicators, the MEPLUS database and presentation module as well as a decision support tool (USE-IT) with which the results of policy options can be evaluated for their social, environmental and economic effects (Lautso, 2002). As a consequence of its area independence, it has been and can be applied in different locations for assessing options for urban sustainability policies in a systematic way. Its application for three test cities (Helsinki, Bilbao, and Naples) shows that the level of sustainability can be increased in a cost-effective way by using a variety of policies, especially pricing and regulatory policies and their combinations. Combining policies can mitigate the negative side effects of individual policies. Fig. 2 shows both the main modules of the SPARTACUS system and a sample outcome of its noise propagation and pollutant dispersion models applied also to the Helsinki Metropolitan Area in Finland. The basic idea behind SPARTACUS is that land

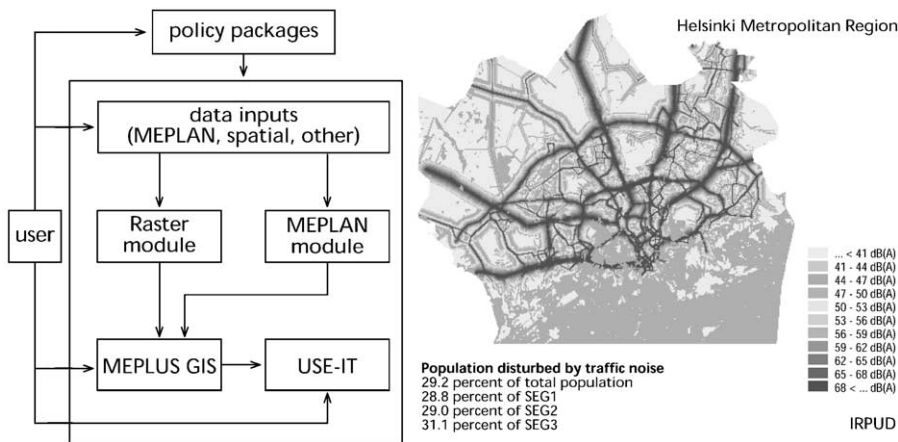


Fig. 2. *SPARTACUS* system modules (left) and sample noise model output for the Helsinki Metropolitan Area (right).

uses, transport and the environment are inextricably linked and cannot be planned in isolation. Therefore, the MEPLAN model brings together the relevant information on transport, land use and economic activity. Development strategies can then be simulated through the model by changing the inputs in order to represent a proposed change in the real world. The model outputs then provide an indication of the wider socio-economic and environmental effects of introducing the change and these effects can be evaluated against an alternative strategy or a 'do-nothing' scenario (<http://www.ltcon.fi/spartacus/default.htm>).

4.2. Capabilities

As with the aims, the capabilities of PSS in the inventory show enormous variation. Some are dedicated to support modelling activities for future population distributions or land-use patterns; some provide tools to support the sketching of new spatial structures; and others are designed purely to allow the visualization of potential spatial developments in all kinds of different ways. Two systems, *GAME* and *K2vI*, can be used to exemplify this diversity.

The New Jersey Growth Allocation Model, known as *GAME*, is a PSS designed for interactive use that identifies the implications of certain trend-based or plan-based scenarios on land development. It has been developed by the New Jersey Office of State Planning (Reilly, 2002) and it allows users to explore various land-use policies in light of both the existing conditions in their town or region and the values and goals of the user. Thus, the model takes user-selected policy choices and forecasts how these policies would affect the location of growth (or decline) and the costs associated with the resulting land-use pattern. Fig. 3 shows schematically the iterative process of exploration of various land-use policies with the help of *GAME*. Based on a regional forecast of relative municipal growth within the region and on a

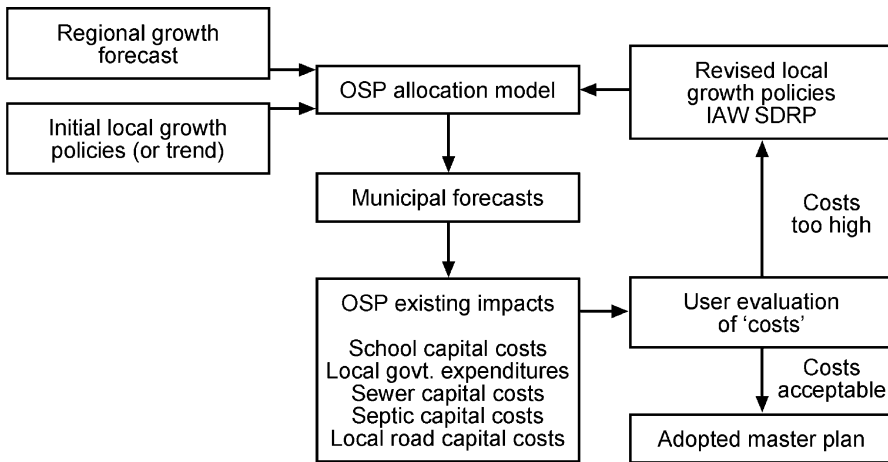


Fig. 3. The iterative process of exploring land-use policies with the help of *GAME*.

user-defined policy plan scenario, *GAME* produces municipality growth estimates and forecasts a range of variables at the municipality scale, like mean household income and average household size. Moreover, it forecasts various impacts associated with each scenario, like capital costs associated with public school buildings and public sewer facilities costs. Using *GAME*, the user can learn if various land-use policies achieve their growth management objectives and identify some of the associated cost impacts of the resulting land-use pattern. Through the iterative process of trial, evaluation and policy modification, the user discovers a preferred set of land-use development policies that produce a land development pattern and its associated costs, which are acceptable and compatible with the goals and values of the person or group.

K2vI (Key to Virtual Insight), on the other hand, is a PSS that allows users to visualise, manipulate and analyse two-dimensional and three-dimensional spatial data within a virtual reality (VR) environment. It was developed initially in the Departments of Geodesy and Technical Informatics at Delft University of Technology (DUT) in co-operation with several engineering companies and the ITC (International Institute for Aerospace Survey and Earth Sciences). Since 1999, *K2vI* has been further developed for urban, military, sports and forestry applications in a joint effort involving Asset Information Systems in New Zealand and the DUT. The software uses the multi-view mode approach where each view mode has a preferred VR display system. Of particular interest is the fact that *K2vI* allows the direct interaction between two-dimensional and three-dimensional GIS and computer-aided design (CAD) data from within the VR environment. In that way, spatial objects can be analysed, manipulated and visualized directly in two and three dimensions without bothering about conversion matters. Fig. 4 illustrates *K2vI* by providing three different views of the centre of the city of Auckland in New Zealand (<http://www.k2vi.com/>).



Fig. 4. K2vI interface showing three different views of the centre of Auckland, New Zealand.

4.3. Content

We have already established that a PSS is a ‘toolbox’ containing various components including data sets, (meta-) information, storage and query tools, analysis methods, theories, indicators, etc., as exemplified by the PSSD. Other PSS, however, are much more specialised and contain only very specific software components to perform specific tasks. Two examples to illustrate specialised PSS are *MIGMOD* and *BIPC model*.

MIGMOD is an example of a PSS that deals with strategic planning support at a national level. It is a prototype internal migration modelling system that was built for the Department of Transport, Local Government and the Regions (DTLR) in the UK by a consortium of researchers based primarily at the Universities of Newcastle and Leeds (Champion, Bramley, Fotheringham, Macgill, & Rees, 2002; Office of the Deputy Prime Minister, 2002). Its purpose is to assist in the understanding of the impacts of changes in various indicators (determinants) on the volume and patterns of internal migration within the UK. At the heart of the system is a two-stage spatial interaction modelling system: a generation model of the flows of out-migrants from a set of regions and a distribution model, which allocates the

predicted out-migrants to destination regions. Both models are driven by sets of explanatory variables and calibrated on historical time series data using Poisson regression techniques, before being used for predicting change under different scenarios. Fig. 5 illustrates sample outputs of *MIGMOD*. On the left hand side, is one example of the user interface tables showing baseline model predictions of flows between a selection of regions of male migrants aged 16–19 in 1996–1997; on the right hand side is an example of the interface mapping facilities showing predicted flows of males aged 16–19 from Leeds to the other 98 regions in the system (and an observed flow included for Northern Ireland).

The *BIPC model* refers to the Bulk Infrastructure Potential Cost model and provides a means for the incorporation of bulk engineering services (water, sanitation, and electricity) cost considerations into the strategic planning process during land suitability assessment (Biermann, 2002). It has been developed by the CSIR Building and Construction Technology in South Africa as a PSS for planners and development decision-makers to enhance the integration of land-use and infrastructure planning. The model is based on threshold analysis, which indicates steep rises in the marginal cost curves associated with further development. Potential costs are calculated on the basis of capacity in the existing system and include additional cost factors such as geo-technical, land-use and environmental conditions to further enhance cost accuracy. The output of the model is in the form of potential cost contours, which facilitates the relative comparison of infrastructure costs for different density scenarios. Fig. 6 shows the three constituting elements of the bulk infrastructure potential cost model. It indicates that when the number of people increases within an area, the infrastructure (e.g., pipes, reservoirs) has to expand, which possesses physical dimensions or thresholds (e.g., pipe diameter, reservoir capacity) directly proportional to size of the infrastructure. The relationship between density and threshold dictates whether or not additional infrastructure

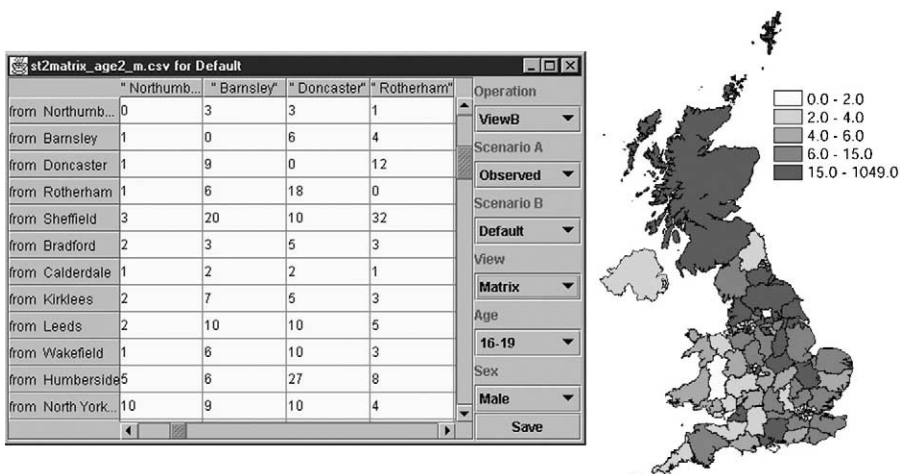


Fig. 5. *MIGMOD* interface showing predictions of out-migration flows in table (left) and map (right).

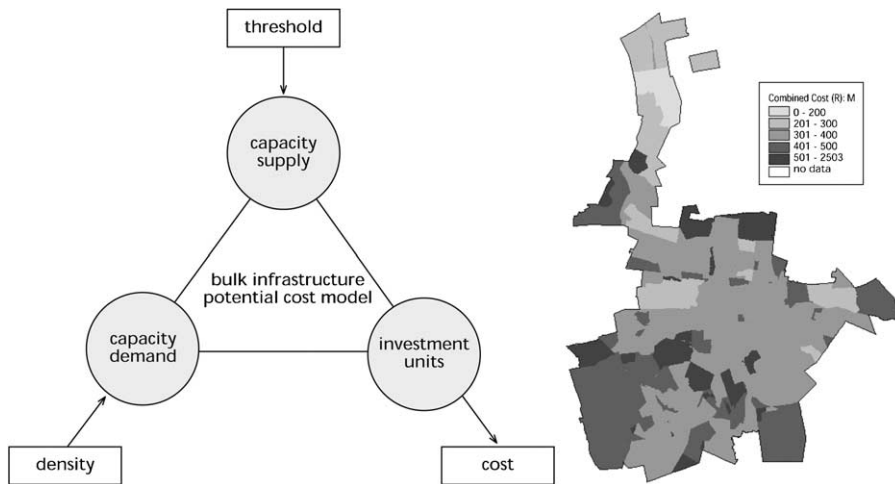


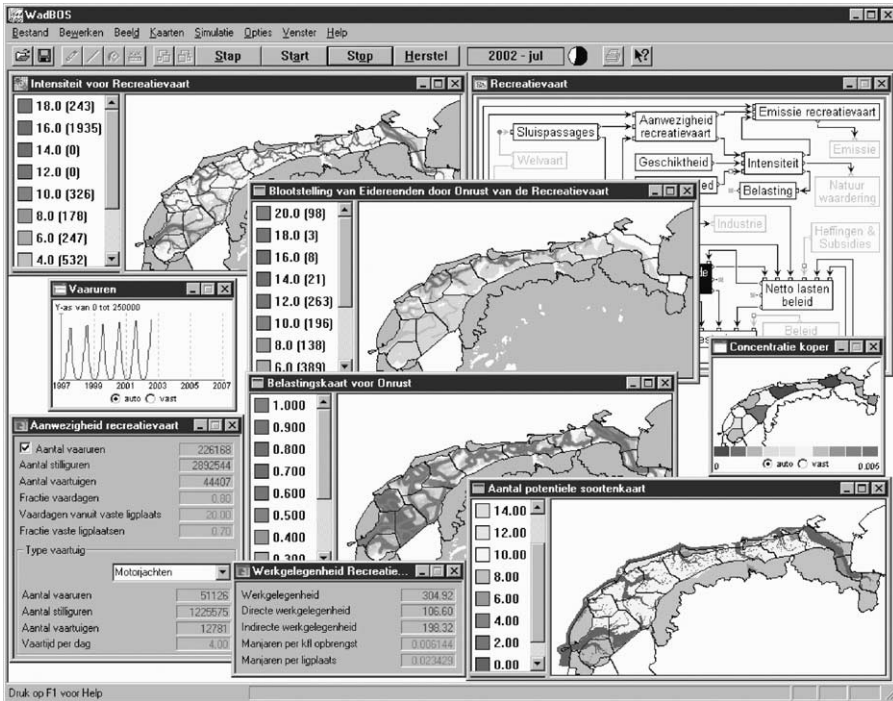
Fig. 6. *BIPC* model (left) and its output based on MCA including bulk infrastructure potential costs (right).

expenditure is necessary to accommodate additional people. In addition, it shows as an example the combined cost output of the *BIPC* model based on a multi-criteria suitability assessment (MCA) including bulk infrastructure potential costs for the city council of Pretoria.

4.4. Structure

PSS differ too in their structure in that some can be considered fully integrated systems in which all components are interconnected closely and others have components that are only loosely connected tools within a container. *WadBos* is an illustration of the former category, *SketchGIS* of the latter.

WadBos is an integrated PSS for the Wadden Sea, an important estuarine system in the northern part of the Netherlands. Although it is considered a protected natural site, it also has important economic functions including, for example, fishing, recreation, transportation and gas exploitation. The management of the different activities and functions of the area are distributed over a great number of institutions, ranging from the local to the European. In order to streamline the process of decision-making, *WadBos* was developed as a highly integrated model representing the ecological and the economic functions of the sea (Engelen, 2000). The component sub-models of *WadBOS* represent processes operating at different time scales and at different spatial scales. *WadBOS* allows a user to assess the effects of policy interventions on the system, to provide insights into both individual domains and the linkages in between the individual domains; to visualize the anticipated policy effects, and to gather and link knowledge concerning the questions at stake. Fig. 7 illustrates *WadBOS*, showing in different visual formats (texts, graphs, maps) the



disturbing consequences of recreational boating on the Wadden Sea for different spatial and time scales (<http://www.netcoast.nl/tools/rikz/WADBOS.htm>).

SketchGIS is a stand-alone toolbox developed to support the first phase of a participatory plan-making process, the creation and evaluation of spatial scenarios for the future. It has been developed by a consortium of public and private sector organisations in the Netherlands and it incorporates a diversity of loosely coupled tools with which associated tasks can be performed: brainstorming sessions can be facilitated by offering tools to identify and store relationships between interesting topics (concept mapping); area-specific data and information in different formats (text documents, HTML-pages, photographic material, three-dimensional figures, etc.) can be handled and presented; the process of sketching spatial scenarios (designing areas for new housing or industry, for example) can be supported; these sketched scenarios can be evaluated with the help of a MCA tool; and the sketched scenarios can be presented in a two- and three-dimensional way (Geertman, 2002a). With the help of the *SketchGIS* toolbox participants within a collaborative planning session can design and discuss expectable, potential and/or desirable futures for their own living area. Fig. 8 illustrates the design tool as part of the *SketchGIS* toolbox with the help of which spatial scenarios can be sketched freely on a background map. It shows the area of the so-called Randmeren, located in the centre of the

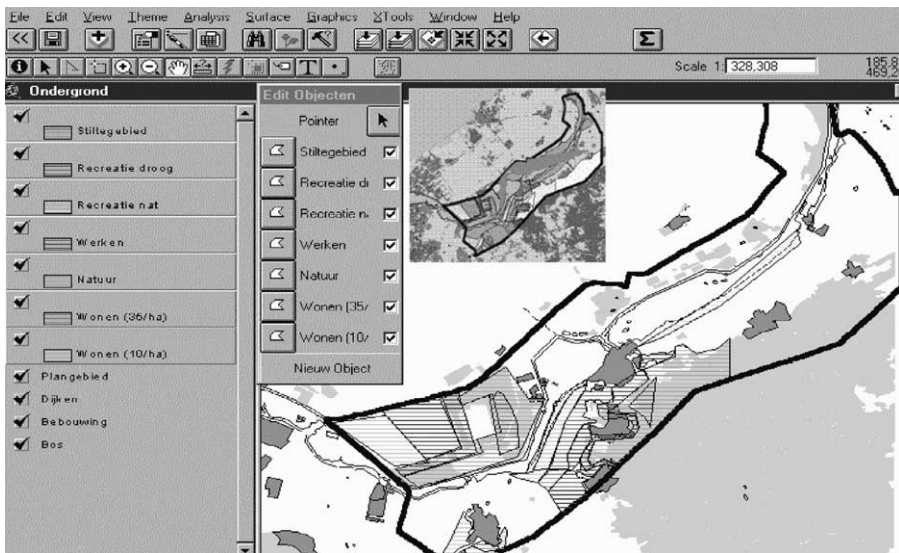


Fig. 8. SketchGIS toolbox for interactive and participatory spatial scenario building.

Netherlands, for which a scenario is generated wherein areas are identified and sketched, suitable for transformation to other land-use categories.

4.5. Technology

In the case of applied technology, some PSS are stand-alone programs while others are developed solely for the Intranet or Internet. Two examples to illustrate this diversity are *What if?*TM and *WBBRIS*. The former is a stand-alone policy-oriented planning tool that can be used to determine *what* will happen *if* certain policy choices are made (e.g. spatial restriction on urbanization growth) and assumptions concerning the future (e.g. population growth rates) prove to be correct (Klosterman, 1999b). It projects future land-use patterns by balancing the supply of, and demand for, land suitable for different uses at different locations, subject to user-specified policies for allocating the projected demand to suitable locations. Alternative visions for an area's future can be explored by defining alternative suitability, growth and allocation scenarios. Fig. 9 illustrates *What if?*TM by showing projected land use scenarios (<http://www.what-if-pss.com/>). It shows the morphological picture of land use in Medina County, Ohio, US, in which the left hand picture indicates the expected land-use configuration for six land-use types for 2045 under conditions of restricted growth controls. In contrast, the right hand picture shows the expected land use configuration for six land-use types for 2045 without restricted growth controls, leading to a very dispersed pattern of residential land use.

In contrast to the stand-alone *What if?*TM PSS, an Internet-based PSS has been developed for the Wide Bay-Burnett region (*WBBRIS, the Wide Bay-Burnett Regional Information System*) of Queensland, Australia (Pettit, Shyy, & Stimson,

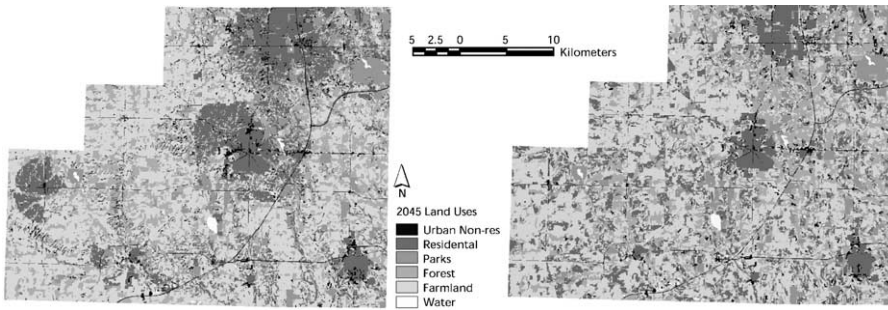


Fig. 9. Output from *What if?*TM showing restricted and unrestricted land use growth control scenarios in Medina County, OH, US.

2002) that incorporates a suite of SDSS tools to undertake multi-scaled planning analysis. For instance, users can perform dynamic analyses of regional demographic and employment trends to formulate thematic maps and associated tabular data results. At the local urban level, users can review the results of a number of ‘what-if’ planning scenarios based on an MCA approach. By providing access to both spatially aggregated and disaggregated socio-economic data, land data and other essential statistical information, linkages between regional and urban scales of analysis can be made such as regional employment shift-share analysis and MCA for local land suitability. Fig. 10 shows the homepage of the *WBBRIS* regional system

Wide Bay-Burnett Regional Information System

This site provides links to a number of decision support tools used to evaluate regional socio-economic growth and decline in the Wide-Bay Burnett Region in Queensland, Australia. Also results of formulated ‘what-if’ planning scenarios are accessible via the links contained below.

Decision Support Tools

- [Shift-Share Analysis](#)
- [Demographic Analysis](#)
- [Multiple Criteria Evaluation](#)
- [Visual Assessment](#)
- [Location Quotient Analysis](#)
- [Input-Output Analysis](#)

What-If Planning Scenarios

- [‘What-If’ Overlay Approach](#)
- [Hervey Bay - Economic Based Urban Growth Scenario](#)
- [Hervey Bay - Social Based Urban Growth Scenario](#)
- [Hervey Bay - Environmental Based Urban Growth Scenario](#)

**Wide Bay-Burnett Region
(Queensland)**

 This on-line resource is based on work in progress being carried out through the School of Geography, Planning and Architecture at the [University of Queensland](#), and is funded through the Australian Research Council SPIRT program in collaboration with the Queensland State Government - [Department of Communication and Information, Local Government, Planning and Sport](#).

Email: [Christopher Pettit](#) or [Tung-Kai Shyy](#)

Fig. 10. Wide Bay-Burnett Regional Information System (*WBBRIS*) homepage.

(http://www.ahuri.uq.edu.au/wide_bay/default.htm). The area being studied is identified, accompanied by several tools like shift-share analysis and input–output analysis, and the various scenarios (economic; social; environmental) that can be applied. Specific to the *WBBRIS* system architecture is the fact that remote users can login the system and receive not just static information (HTML pages) but also dynamic information (produced at distance and on demand by an Internet GIS server) concerning the topic requested.

The examples that we have selected in this section provide insights into the diversity of PSS that can be found in contemporary planning practice. In addition to the criteria that we have used to summarize the PSS, other criteria can be identified such as age (some PSS have existed for over 10 years now), professionalism (some are in a stage of prototyping while others are fully-grown products), stage of commercialisation (some are mature off-the-shelf commercial products, others are available free of charge) or targeted populations (stakeholders, policy-makers, planners, wider public). Assessment of the utility of the PSS contained in the inventory for actual planning practice still has to be fully and properly evaluated. This however, will require a longer period for practitioners to work with these systems and to gain experience with their practical application. The recommendations in the next section, with which we conclude the paper, can be of help in this evaluation.

5. Recommendations

Based on the relatively small number of real world experiences with PSS, some recommendations can be made about the future of PSS development and application. First, PSS should be an integral part of the planning process and context. The Shaping Dane Project website (<http://www.lic.wisc.edu/shapingdane/>) is an excellent example of where different planning and participatory activities are linked to each of the planning phases. For participants, it is important to know at which phase of the planning process one can perform each particular kind of task or activity. A consequence of this is that PSS should be attuned to the specific characteristics of the planning process and context at hand. Each planning process is subject to certain time constraints, allows for a particular degree of participation, involves a certain amount of publicity, etc., and these factors influence the potential for and restrictions on the use of the technology. Likewise, the characteristics of the policy context (e.g., democracy or autocracy) influence the preferred technology (e.g., transparent or black-box) and the way in which it is used. For instance, a participatory planning process and context defines a very different set of criteria with which the PSS must conform in comparison to the classical planning process and context in which just planning professionals make use of the PSS. The differences involved relate to characteristics such as interactivity, flexibility, adaptability, maintainability and scale dependence.

Second, related to the first recommendation, PSS should meet user and context requirements too, besides the conformation to requirements of the planning process and context. For example, PSS should be developed that reflect the knowledge and

skills of their direct applicants, which may involve meeting multiple levels of expertise. They should be attuned to the professional tasks required at a certain phase in the planning process (sketching, modelling, impact assessment and evaluation) and to the capabilities of the audience with whom communication is required, i.e., knowledge and skill levels of lay persons and/or stakeholders as well as more technical users. Obviously, these user and context requirements will be very different and probably much more difficult to meet within a participatory planning process than in a classical planning process. The diversity of knowledge and skill levels of users within participatory planning processes will put tighter demands on the PSS to communicate properly the relevant planning information.

Third, one should be aware of the fact that people address issues from an interdisciplinary perspective. In contrast to science in which the study of reality has been categorized in separated disciplines, laypersons do not tend to think in accordance with distinct disciplines but in terms of integrated systems or parts thereof. In practice, people encounter problems and ask for solutions and usually neither the problems nor the solutions will be confined to the artificial boundaries of disciplines. As a consequence, PSS that intend to connect to peoples' way of thinking should address issues in an interdisciplinary manner, linking the spatial to the social, the environmental to the economic and so forth.

Fourth, PSS should take seriously its users and leave them with the feeling that they have been taken seriously. Although this sounds very much like a statement of the obvious, experience suggests that this is not always the case. For example, the population of one local community in the USA have participated actively in planning sessions in which environmentally sensitive futures have been designed for their region, whilst at the same time, the Governor of the State in which their region is located and who has claimed to be a strong protagonist of participative sessions, is making secret deals with Federal Government for the construction of a large nuclear waste disposal site in that region.

Fifth, the user-interface of the PSS should be sensitive to the characteristics of the user, to the kind of information that it communicates to that user, and to the types of intended use that is made of the information provided. For instance, in general one might presume that people are highly visual in their requirements for communication, thinking and forms of expression. Consequently, it might be assumed that imagery such as photos, three-dimensional images, architectural drawings, cardboard replicas and VR-movies, would be highly effective in the understanding and communication of different visions for the future, and much more so than text statements about different scenarios. However, in particular instances, these visualisations can be highly inappropriate and even undesirable in communication processes, for instance, when these visualisations distort reality purposefully to achieve selective or biased interests.

Sixth, obviously the PSS should be focused in particular on the planning problem at hand. For some strategically oriented planning tasks, this means the incorporation of tools for sketching, modelling and impact analysis. The activities of detecting the most likely future, exploring potential scenarios, and/or designing desirable futures belong to the core of strategic planning tasks. In that, of particular

interest is the question of acceptance of the inputs (such as reliable, up-to-date data) and outputs (such as meaningful modelling results) of a PSS. In general, transparency is seen as one of the most important prerequisites for support tools and for the way in which they are handled.

Finally, PSS should be appealing; they should fulfil participants' needs and wishes, and allow the participants to enjoy using them too. Especially in circumstances in which people are expected to invest their free time to participate in planning sessions, one should be aware of the sensitivities associated with this investment of time and energy. Various means can be used to ensure that PSS encourage participation, including the incorporation of some type of competition (with prizes available) or the establishment of resource centres as places equipped with decision/planning support tools to structure and promote dialogue and community issues.

In summary, we can conclude that the application of PSS is currently still in its infancy and there does not yet exist a reservoir of real-world experience with PSS with which to conduct a detailed evaluation. However, there is little doubt that the development of PSS is on an upward trajectory, that new and exciting tools are being created and applied in different parts of the world and that future innovations will be stimulated as the awareness of contemporary systems, as exemplified in this paper, becomes more widespread in planning practice.

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